MILITARY SPECIFICATION

PANEL ASSEMBLY, RELAY, FOR SEARCHLIGHT, INFRARED, 28-VOLT, 110-AMPERE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the panel assembly, relay for searchlight, infrared, 28-volt, 110-ampere.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive, Adhesive, Filament Reinforced

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MIL-P-116	Preservation-Packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40 x 48 Inch Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-613844	Panel Assembly, Relay

(Copies of specifications, standards, and drawings required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y32.16-1975	Reference Designations for Electrical and Electronic Parts and Equipments

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

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3. REQUIREMENTS

3.1 Description. The panel assembly, relay, is a solid state electronic device consisting of electronic timing and switching circuitry and hereinafter referred to as the relay panel.

3.2 Construction. The relay panel shall be constructed in accordance with SC-DL-613844 and as specified herein.

3.3 First article. The supplier shall furnish and test assemblies as specified in the contract or purchase order (see 6.2).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Operating and environmental.

3.5.1 Operation. The relay panel shall automatically time two circuits, one of which will be for a duration of 2.5 seconds, plus or minus 0.5 second, and the other for a duration of 17.5 seconds, plus or minus 2.5 seconds. The relay panel shall also contain switching relays and circuitry to apply mode change power to external motors. The relay panel shall operate with an input voltage from 22 to 28 volts dc.

3.5.2 Vibration. The relay panel shall not be damaged (see 6.3.1) by vibration throughout a frequency range of 15 to 120 hertz (Hz) at 2.5 g's, ± 0.5 g. (see 6.3.2).

3.5.3 Low temperature storage. The relay panel shall not be damaged by storage at an ambient temperature of minus 62°C with relative humidity uncontrolled.

3.5.4 High temperature operation. The relay panel shall operate as specified herein at any ambient temperature from plus 23°C to plus 93°C.

3.5.5 Low temperature operation. The relay panel shall operate as specified herein at any ambient temperature from plus 23°C to minus 54°C with humidity uncontrolled.

3.5.6 Thermal shock. The relay panel shall not be damaged by temperature shock from minus 54°C to plus 93°C with humidity uncontrolled.

3.6 Life. The relay panel shall operate as specified herein, a minimum of 10,000 consecutive times without failure.

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3.7 Treatment and painting. Treatment and painting shall be in accordance with MIL-STD-171 as prescribed by the applicable drawing.

3.8 Marking. Unless otherwise specified the relay panel and all components shall be marked and identified per MIL-M-13231 and ANSI Y32.16 1975.

3.9 Workmanship. Unless otherwise specified, workmanship shall conform to Requirement 9 of MIL-STD-454.

3.9.1 Soldering. Unless otherwise specified, soldering shall conform to Requirement 5 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified by the contract purchase or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services comform to prescribed requirements.

4.1.1 Parts and materials inspection. The supplier is responsible for insuring that parts and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as specified herein.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.5).

c. Inspection of preparation for delivery (4.8).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 Inspection. Each first article shall be inspected in accordance with Table I. Presence of one or more defects shall be cause for rejection of that relay panel.

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TABLE I Inspections

Defects	Requirement paragraph	Inspection criteria
Workmanship not as specified	3.9	MIL-STD-454
Treatment and painting not as specified	3.7	MIL-STD-171
Markings not as specified	3.8	MIL-M-13231 and ANSI Y32.16 1975

4.3.2 Tests. Following successful completion of the inspections specified in 4.3.1, each first article relay panel shall be subjected to all inspections in Table II. Failure of any inspection shall be cause for rejection of that relay panel. Inspections shall be performed in the order listed in Table II.

4.4 Inspections covered by subsidiary documents. The relay panel shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order (see 4.3.1 and Table I).

TABLE II First article inspection

Inspection	Requirement paragraph	Test paragraph
Operation	3.5.1	4.6.1
Vibration	3.5.2	4.6.2
Low temperature storage	3.5.3	4.6.3
High temperature operation	3.5.4	4.6.4
Low temperature operation	3.5.5	4.6.3
Thermal shock	3.5.6	4.6.4
Life	3.6	4.7

4.5 Quality conformance inspection.

4.5.1 Group A Inspection. Each relay panel shall be tested in accordance with Table III. Failure of any test shall be cause for rejection of that relay panel.

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TABLE III Group A Inspection

Inspection	Requirement paragraph	Test paragraph
Operation	3.5.1	4.6.1

4.5.2 Group B Inspection. Group B inspections shall be conducted on relay panels selected from units which passed the examination and test in 4.3.1 and 4.5.1. Sampling shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level I combined. Tests shall be conducted in accordance with Table IV.

TABLE IV Group B Inspection

Inspection	Requirement paragraph	Test paragraph
Vibration	3.5.2	4.6.2
High temperature operation	3.5.4	4.6.4
Thermal shock	3.5.6	4.6.4
Low temperature storage	3.5.3	4.6.3
Low temperature operation	3.5.5	4.6.3

4.5.3 Group C Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A inspection. Sample units shall be selected in accordance with 4.5.4.

4.5.4 Sampling for Group C inspection. Three sample relay panels shall be slected at random for inspection in Table V from each 100 relay panels produced. The first sample selected shall be at the start of the contract from the first quality conformance inspection lot.

TABLE V Group C Inspection

Inspection	Requirement paragraph	Test paragraph
Life	3.6	4.7

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4.5.5 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2).

4.5.6 Reinspection of Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Test methods. All tests shall be conducted at room temperature (see 6.3.3) unless otherwise specified in the individual test procedure. Test facilities, chambers and apparatus used in conducting tests specified shall meet the requirements of MIL-STD-810.

4.6.1 Operation. Connect the relay panel to a test fixture in accordance with Figure 1 and perform the following test:

a. Connect the fixture to a 22 $\pm$ 0.5 volt dc power supply.

b. Turn S 1 to "ON" left and right momentarily. DS 1 shall indicate in both positions.

c. Turn S 2 to "ON" left and right momentarily. DS 2 shall indicate in both positions.

d. Turn S 3 to "ON" left and right momentarily. DS 3 shall indicate in both positions.

e. Turn S 4 to "ON" momentarily. DS 4 shall indicate for 17.5 seconds, plus or minus 2.5 seconds.

f. Turn S 5 to "ON". DS 5 shall indicate for 2.5 seconds, plus or minus 0.5 second. Turn S 5 to "OFF".

g. Repeat b. thru f. at 28 $\pm$ 0.5 volts dc.

h. Failure of the relay panel to perform any of the above indicated operations shall constitute failure of this test.

4.6.2 Vibration. Mount the relay panel rigidly to a vibration table by the 4 mounting points. Perform 4.6.1. Vibrate the relay panel in each of 3 mutually perpendicular planes for 15 minutes each at an amplitude of 0.06 inch, plus or minus 10 percent total excursion, while cycling the frequency from 10 to 120 to 10 Hz. Perform 4.6.1. Failure to operate as specified in 4.6.1 or any evidence of damage shall constitute failure of this test.

4.6.3 Low temperature storage and operation. The relay panel shall be placed in a test chamber and exposed to an ambient temperature of minus 62°C, $\pm$ 3°C for a period of 4 hours. The test chamber temperature shall be raised to minus 54°C, $\pm$ 3°C. After 4 hours at minus 54°C the relay

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panel shall be tested in accordance with 4.6.1. Failure of the relay panel to operate as specified in 4.6.1 or evidence of damage shall constitute failure of this test.

4.6.4 High temperature operation and thermal shock. The relay panel shall be placed in the test chamber and subjected to 2 hours at plus 93°C, ± 3°C. Within five minutes the relay panel shall be placed in a test chamber stabilized at minus 54°C, ± 3°C for two hours. Within five minutes the relay panel shall be placed in a test chamber stabilized at plus 93°C, ± 3°C. After two hours at plus 93°C and while the relay panel is in the plus 93°C environment the relay panel shall be tested in accordance with 4.6.1. Failure of the relay panel to operate as specified in 4.6.1 or evidence of damage shall constitute failure of the test.

4.7 Life. Perform 4.6.1 and record the results. Connect the relay panel to a tester which performs all of the testing functions as specified in 4.6.1 (this tester may be automated). Operate the relay panel at 28, ± 0.5 volts dc for 10,000 consecutive cycles. Each cycle shall consist of one operation as specified in 4.6.1, except that the timing cycles need not be measured and that only one voltage need be checked per cycle. Provision shall be made to automatically count the number of cycles completed. At the conclusion of 10,000 cycles as specified above, perform 4.6.1 and record the results. Failure of the relay panel to operate as specified in 4.6.1 shall constitute failure of this test.

4.8 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each panel assembly, relay shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. Non required.

5.1.1.4 Unit packaging. Each panel assembly, relay shall be packaged method III in accordance with MIL-P-116 as follows: Cushion each panel assembly on all surfaces with cells and pads or both fabricated from fiberboard conforming to PPP-F-320, Type CF, class weather-resistant, grade W6c, designed to absorb the shock of impact in handling and transit. Place the cushioned panel assembly within a close-fitting fiberboard box conforming to PP-B-636, W6c. Close the box in accordance with the appendix of the box specification.

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5.1.2 Level C. Panel assembly, relay shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2).

5.2.1 Level A .

5.2.1.1 Consolidation. A quantity of panel assembly, relay, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, Type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. A quantity of containers, packed as specified in 5.2.1.1 shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B.

5.2.2.1 Consolidation. A quantity of panel assembly, relay, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

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5.2.2.2 Palletized load. A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C.

5.2.3.1 Consolidation. A quantity of panel assembly, relay, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The relay panel is intended for use in searchlights, infrared, conforming to MIL-S-52444.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation and packaging and the level of packing required (see 5.1 and 5.2).

c. Actions required relative to Group C failure (see 4.5.5).

d. Schedule for First Article Testing (see 3.3).

6.3 Definitions. For the purpose of this specification, the following definitions apply.

6.3.1 Damage . Damage is defined as deformation, loosening, breakage, corrosion, or change of fit of any component or part.

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6.3.2 "g". g is defined as an acceleration or deceleration of 32.17 feet per second per second.

6.3.3 Room temperature. Room temperature is defined as 23°C plus or minus 3°C.

CUSTODIAN:

Army-EL
Navy-EC
Air Force - 84

USER ACTIVITY:

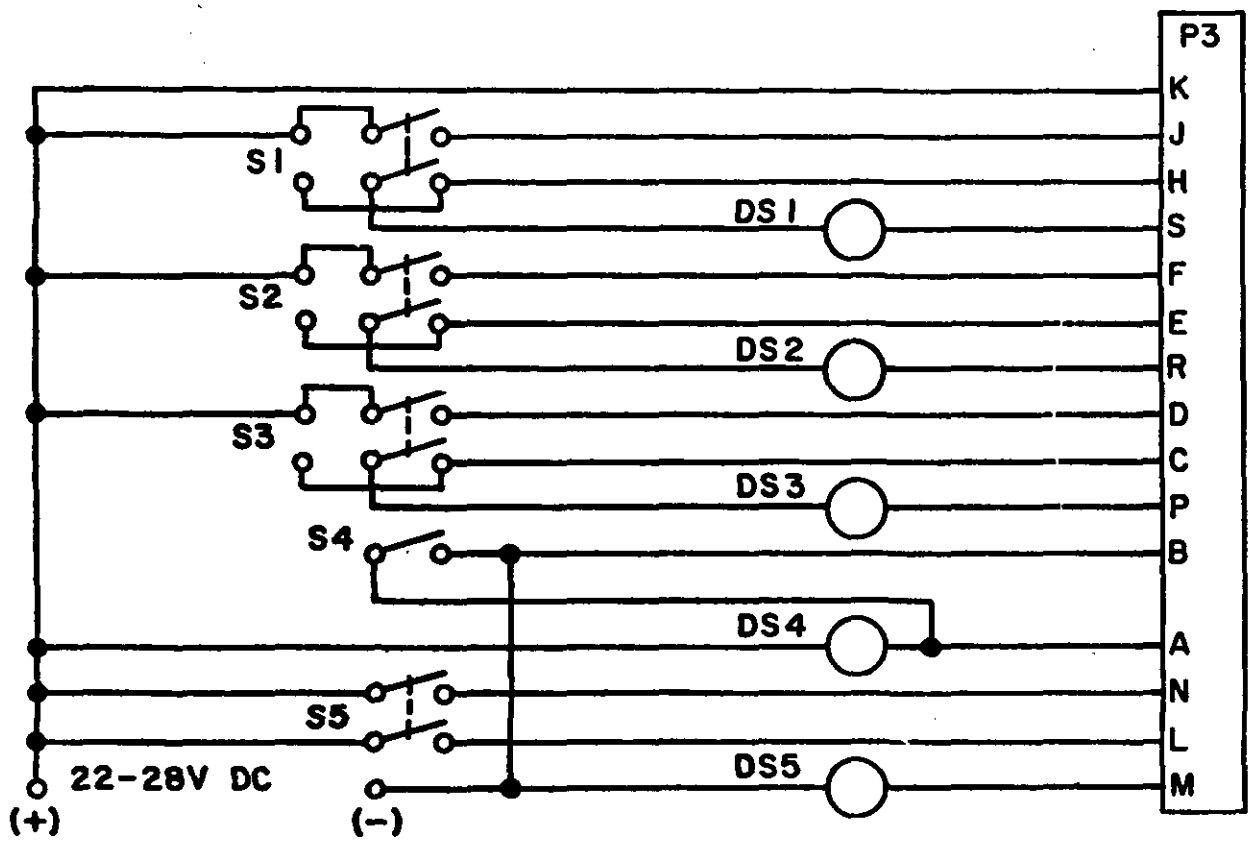
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PREPARING ACTIVITY:

Army-EL

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NOTES:

1. S1, S2, AND S3 ARE DPDT MOMENTARY SWITCHES.
2. S4 IS A SPST MOMENTARY SWITCH.
3. S5 IS A DPST SWITCH.
4. P3 CONFORMS TO MS24011-1. P3 SHALL BE MATED TO J3 (P/N SC-C-634241) OF THE RELAY PANEL ASSEMBLY DURING TEST.
5. DS1 THRU DS5 ARE 28 VOLT, 0.17 AMP. LAMPS.

FIGURE 1 TEST FIXTURE SCHEMATIC

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